



Customer Behavior Analysis

Unlocking insights from retail data to drive revenue growth and customer retention

About the Dataset

Retail/E-commerce

Structured tabular data at customer transaction level

Data Mix

Numerical: Age, Purchase Amount, Previous Purchases

Categorical: Category, Payment Method, Season

Purpose

Customer behavior analysis and business decision-making

Why This Data Matters



Business Value

- Identify high-value customers
- Understand purchase patterns
- Optimize discount strategies
- Improve marketing campaigns
- Increase customer lifetime value

Breaking Down the Problem

01

Low Customer Retention

Customers not purchasing frequently enough

03

Product Demand

Which categories and products perform best

05

Seasonal Variation

How seasons impact sales patterns

02

Ineffective Marketing

Discounts and promo codes may not be working

04

Payment Insights

Preferred payment methods and trends

06

Customer Segmentation

Identify high-value, frequent, and discount-sensitive customers

Data Dictionary

Key columns used for analysis

Column	Description
Customer ID	Unique identifier for each customer
Age, Gender	Demographic information
Item Purchased	Specific product purchased
Category	Product category (Clothing, Electronics, etc.)
Purchase Amount	Amount spent by the customer
Location	Customer's geographic location
Season	Season of purchase (Winter, Summer, etc.)
Review Rating	Customer rating for the product
Subscription Status	Whether customer is subscribed
Shipping Type	Type of shipping selected
Discount Applied	Whether discount was applied
Previous Purchases	Number of past purchases
Payment Method	Mode of payment
Frequency of Purchases	Purchase frequency (Weekly, Monthly, etc.)

Data Cleaning Process

1

Data Loading

Imported using Pandas and verified structure

2

Consistency Check

Fixed category inconsistencies with mapping dictionary

3

Missing Values

Handled Size, Review Rating, Purchase Amount with domain logic

4

Duplicates

Removed duplicate Customer IDs

5

Export

Standardized columns and exported to PostgreSQL

Key Findings: Revenue & Categories

\$486K

Electronics

Highest revenue category

\$195K

Accessories

Second highest performer

\$183K

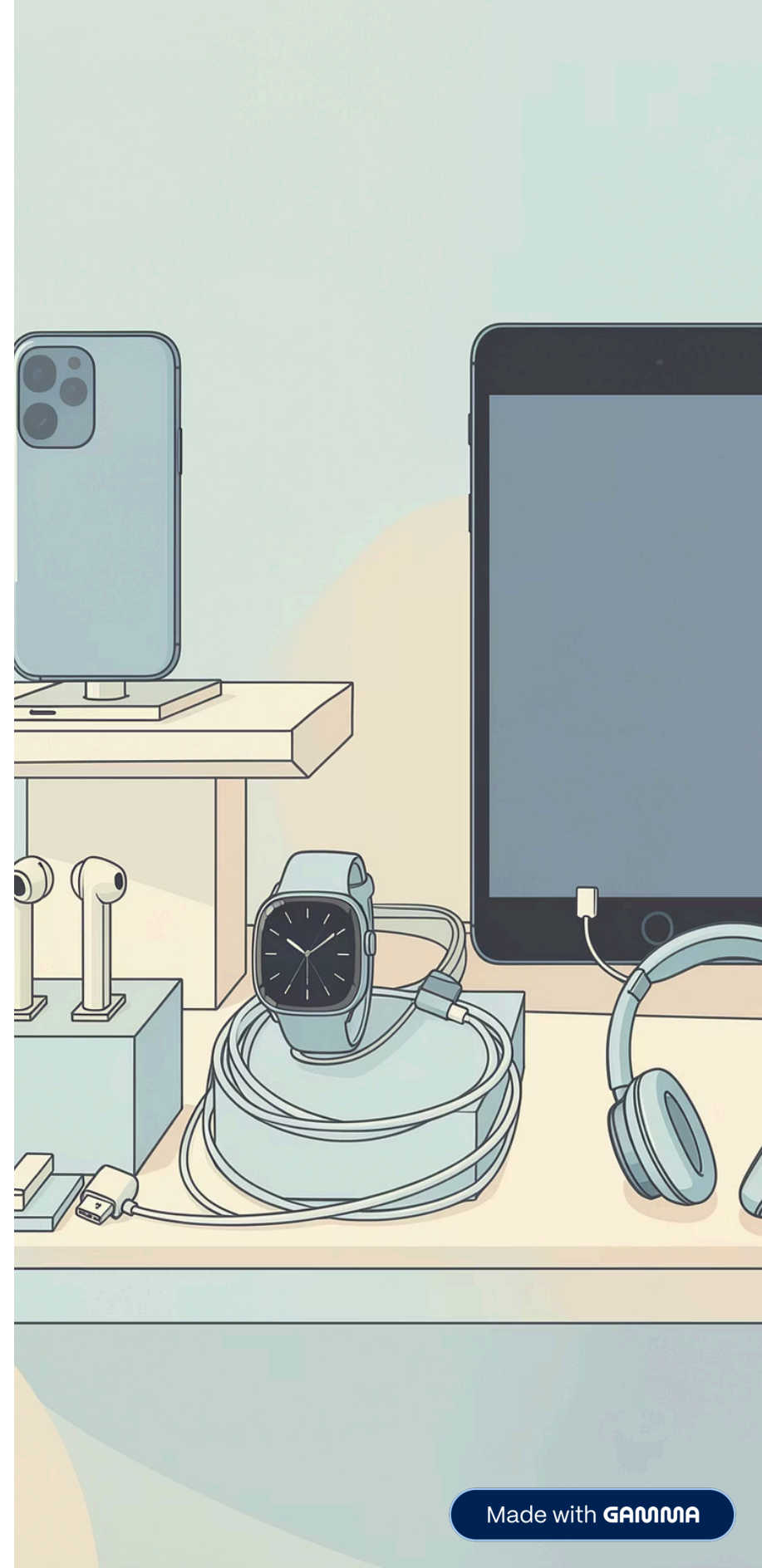
Clothing

Third in revenue

\$111K

Footwear

Fourth in revenue



Discount Strategy Insights

Discount Impact

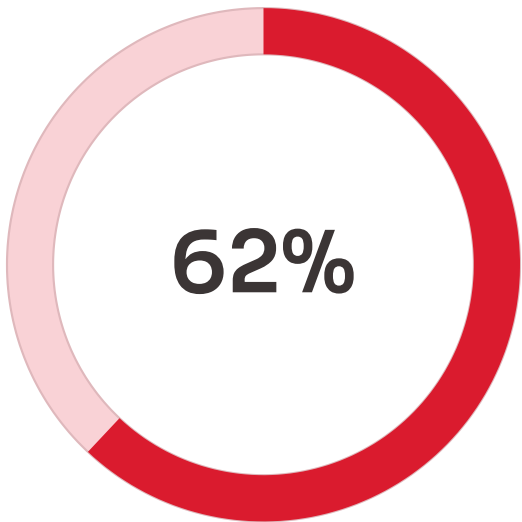
Discounts increase average purchase from \$182 to \$219, but total revenue slightly lower (\$492K vs \$502K without discounts)

Top Discount Products

- Laptop: 51.75% discount usage
- Phone: 51.46% discount usage
- Headphones: 51.20% discount usage

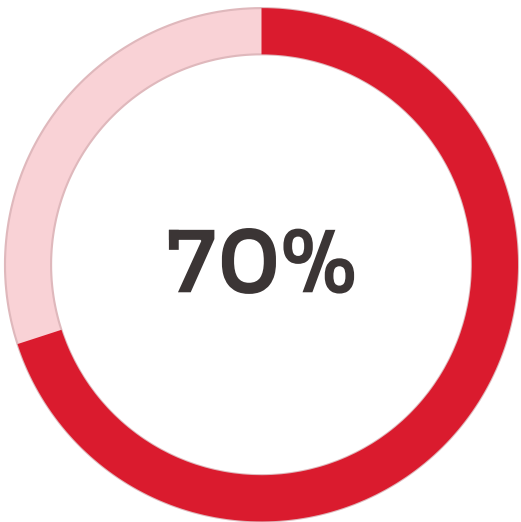


Customer Segmentation



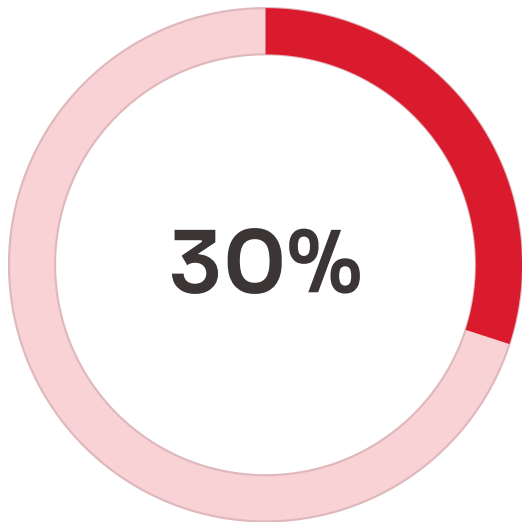
Normal Buyers

Not subscribed (658 customers)



Repeat Buyers

Not subscribed (2,763 customers)



Repeat Buyers

Subscribed (1,190 customers)

Age Groups

51+ age group generates highest revenue (\$399K), followed by 35-50 (\$267K)

Payment Methods

Debit Card leads (\$250K), followed by Cash and Credit Card (\$230K each)

Power BI Dashboard Highlights



5,000 Customers

Total unique customers in dataset



30 Products

Unique items across all categories



3.62 Rating

Average customer review score



\$994K Spend

Total revenue generated



\$199 Average

Per customer spending